EE-I-1699
January 14, 1972
(See 6.6)

FEDERAL SPECIFICATION

ICE CREAM MIXES, REGULAR AND IMITATION, DEHYDRATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers non-perishable ice cream and imitation ice cream mixes for use by all Federal agencies.

1.2 Classification.

Type I - Regular (containing milk fat)
Type II - Imitation (containing no milk fat)

Style A - Containing only coconut fat
Style B - Containing vegetable or animal fat or combinations thereof, other than coconut fat and milk fat

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gauge and Lighter
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)

FSC 8910

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence In
MIL-F-35081	-	Fat, Hydrogenated, Vegetable

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-671	-	Sanitary Standards for Milk Evaporating and Drying Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

U.S. Department of Health, Education, and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers

(Single copies may be obtained from the Division of Food Sanitation, Food and Drug Administration, U.S. Department of Health, Education and Welfare, Washington, D.C. 20402.)

U.S. Department of Agriculture

Visual Aids for Inspection of Metal Containers

Directory of Consumer Protection Programs, Establishments, Circuits, and Officials

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

U.S. Standard for Grades of Nonfat Dry Milk (Spray Process)

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to the Consumer and Marketing Service, Dairy Division, U.S. Department of Agriculture, Washington, D.C. 20250.)

Brucellosis Eradication - Recommended Uniform Methods and Rules, Agricultural Research Service, U.S. Department of Agriculture, Bulletin ARS 91-79

(Copies may be obtained from U.S. Department of Agriculture, Agricultural Research Service, Washington, D.C. 20250.)

Association of Official Analytical Chemists

Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

American Oil Chemists' Society

Official and Tentative Methods

(Application for copies should be addressed to the American Oil Chemists' Society, 35 E. Wacker Drive, Chicago, Ill. 60601.)

American Public Health Association

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W., Washington, D.C. 20036.)

American Dry Milk Institute, Inc.

Standards for Grades of Dry Milks, Including Methods of Analysis Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 N. Franklin St., Chicago, Ill. 60606.)

Munsell Color Co., Inc.

Munsell Color Discs

(Munsell Color Discs may be obtained from the Munsell Color Co., Inc., 2441 No. Calvert St., Baltimore, Maryland 21233.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington D.C. 20036.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Milk. The raw milk used in the preparation of this product shall comply with all the quality requirements for No. 2 (or better) specified in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service. Undergrade milk shall not be used. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(1) A Modified Accredited Tuberculosis Area; and

(2) Either (a) A certified Brucellosis Area; or
(b) Modified Certified Brucellosis Area; or

(3) An area in the process of being certified by the USDA in accordance with USDA Agricultural Research Service Bulletin 91-79. The milk shall be practically free from colostrum, fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.

3.1.2 Skimmed milk, concentrated skimmed milk, and concentrated whole milk. Skimmed milk, concentrated skimmed milk, and concentrated whole milk shall be prepared from raw whole milk conforming to the requirements of 3.1.1.

3.1.3 Cream, plastic cream, churned fat, anhydrous milk fat, and butteroil. Cream, plastic cream, churned fat, anhydrous milk fat, and butteroil shall be prepared from raw whole milk conforming to the requirements of 3.1.1.

3.1.4 Nonfat dry milk. Nonfat dry milk shall comply with the U.S. Standards for Grades of Nonfat Dry Milk (spray process) for Extra Grade, low heat powder.

3.1.5 Modified milk products. Modified milk products shall be of edible grade derived by the treatment of milk solids to adjust lactose, calcium or minerals, shall be prepared from milk conforming to the requirements of 3.1.1 and shall conform to the requirements of Frozen Desserts, Definitions and Standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.1.6 Vegetable fat. Vegetable fat other than coconut fat and milk fat shall conform to the requirements of MIL-F-35081, except that the Wiley melting point shall be 93° ± 3° F.

3.1.7 Coconut fat. Coconut fat shall comply with the following physical and chemical requirements.

3.1.7.1 Physical requirements. Coconut fat shall have a clean bland flavor and odor free from extraneous materials. The liquified fat shall be clear and brilliant. The color shall be 15 yellow and 1.5 red, maximum (Lovibond).

3.1.7.2 Analytical requirements.

Moisture	None
Wiley melting point	93° ± 3° F.
Peroxide value, not higher than	1.0 meq/kg of fat
Free fatty acids (as oleic), not more than	0.05 percent
Iodine value, not higher than	9
Fat stability, (AOM) not less than	100 hours

3.1.8 Sweetening ingredients. Sweetening ingredients shall consist of sucrose with not more than 20 percent substitution of invert sugar, corn sugar, corn sirup or corn sirup solids.

3.1.9 Vanilla flavoring. Vanilla flavoring shall be vanilla powder (sugar), vanillin or combinations thereof.

3.1.10 Stabilizers, stabilizing salts, emulsifiers, and coloring material. Stabilizers, stabilizing salts, emulsifiers, and coloring material shall comply with 3.6 and with the Frozen Desserts, Definitions and Standards.

3.1.11 Salt. Salt shall be white, refined, noniodized sodium chloride.

3.2 Processing.

3.2.1 Raw milk and dairy products. All raw milk and dairy products not processed within two hours after receipt shall be cooled to 45° F. or lower and maintained thereat until processing is started.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.2.2 Pasteurization. The entire liquid basic mix, with the exception of part or all of the sugar and flavoring, shall be pasteurized by heating every particle to at least 155° F. and holding the mix at this temperature for at least 30 minutes or by any other pasteurization process recognized as equally efficient by the U.S. Public Health Service and approved by the appropriate State or local authority.

3.2.3 Drying. The liquid or concentrated mix shall be spray dried. The dried product shall be cooled to 90° F. to 110° F. immediately upon removal from the spray drier and prior to final packaging or bulk storage. Prior to packaging, the sugar and flavoring shall be blended into the product if these ingredients were not added to the liquid mix prior to pasteurization. Spray driers shall be of the continuous product removal type and wet collectors shall not be used.

3.2.4 Prepackaging holding requirements. The finished product may be held for not more than seven days after manufacturing before final packaging. If not packaged in the final container within 24 hours after manufacture, the product shall be continually stored either in an atmosphere of inert gas or under a constant vacuum of not less than 20 inches. If the storage container is opened or the vacuum broken more than 24 hours prior to the final packaging, the above procedure shall be repeated. Dry blending of the product more than 24 hours prior to final packaging shall not be permitted unless the above operations are repeated. The storage containers shall not have surfaces of iron or copper exposed to the product.

3.3 Finished product.

3.3.1 Physical requirements of dehydrated mix.

3.3.1.1 Flavor and odor. There shall be no foreign flavor or odor such as, but not limited to: oxidized, rancid, stale, tallowy, scorched or fishy.

3.3.1.2 Color. The color shall be white or light yellow.

3.3.1.3 Foreign material. There shall be no foreign material such as, but not limited to: dirt, insects or insect parts, wood, paint, glass or metal particles.

3.3.1.4 Lumps. The product shall be free from lumps that do not fall apart under slight pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.3.2 Physical requirements of rehydrated mix and hardened ice cream.

3.3.2.1 Reconstitution of the mix. The mix shall reconstitute readily with potable water to a typical fluid ice cream mix.

3.3.2.2 Freezing performance. The mix, when reconstituted in accordance with 5.4.2.1 and when frozen under suitable commercial conditions, shall have good whipping capacity. Under these specified provisions, the mix shall yield a semi-frozen product having a stiff consistency and relatively dry appearance without affecting the attainment of 100% overrun in less than 12 minutes of freezing (batch method) at a drawing temperature of from 23° to 25° F. The semi-frozen ice cream upon extrusion, shall be characteristically creamy and smooth in texture with a firm, compact body. Continuous frozen ice cream (when applicable) made from the mix shall result in a product of similar (or better) body and texture attributes as for batch frozen ice cream.

3.3.2.3 Hardened ice cream. The hardened ice cream shall have a smooth texture and shall be free from palpable ice or lactose crystals. The flavor shall be pleasing, desirable and typical of vanilla ice cream.

3.3.3 Analytical requirements. The product shall comply with table I.

TABLE I. Analytical requirements

Item		Requirement
	Dehydrated mix	
1	Total fat, not less than	27.40 percent
2	Protein, not less than	9.80 percent
3	Moisture, not more than	2.0 percent
4	Oxygen in head space gas not more than	2.0 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE I. Analytical requirements (cont'd)

Item		Requirement
	Dehydrated mix	
5	Scorched particles not more than	15.0 milligrams/22 gram sample
6	Iron 1/, not more than	10.0 parts per million (p.p.m.)
7	Copper 1/, not more than	1.0 p.p.m.

1/ If a copper and iron test made on the initial production lot complies with the specification requirement, subsequent testing shall be on a reduced basis. Reduced testing shall mean the testing of not less than one randomly selected sample every 6 months. If the initial production lot does not comply with the minimum copper and iron requirement, testing of subsequent lots in each contract shall be continued until two successive lots are in compliance with the requirements.

3.3.4 Microbiological requirements. The dehydrated product shall comply with table II.

TABLE II. Microbiological requirements

Item		Requirement
1	Standard plate count, not more than	30,000 per gram
2	Coliform count, not more than	10 per gram
3	Yeast and mold (combined count) not more than	20 per gram
4	Salmonella test	Negative

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

3.4 Plant qualification. The product shall originate, be processed, handled, packaged and stored in plants which are approved under the USDA General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service, and listed by the U.S. Department of Agriculture in the document titled: Dairy Plants Surveyed and Approved for USDA Grading Service and approved for the finished product described in this specification, or from plants listed with a rating of 90 or more in the document titled 'Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers', or in plants meeting the requirements of MIL-STD-671.

3.5 Delivery of product. The product shall be delivered within 60 days after date of manufacture.

3.6 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records of the examination and tests as specified shall be complete and made available to the Government.

4.2 Sampling. Samples of finished products covered by this specification and samples of any materials, components, or constituents entering products covered by this specification may be taken at any time by the Government inspector to determine compliance with the requirements. Such samples as are required shall be at the expense of the supplier.

4.3 Inspection.

4.3.1 General. Inspection shall be made by the procuring agency, or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any point in transit, or after delivery at the point of destination (see 6.7).

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:43Z
Check the source to verify that this is the current version before use.

4.3.2 Plant qualification conditions.

4.3.2.1 Civil agencies. Inspection of plants producing the product will be conducted upon request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments which maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.3.2.2 Military agencies. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification of 3.4.

4.4 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Inspection of milk and milk products.

4.4.1.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in disease free herds as specified in 3.1.1. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the Agricultural Research Service of the Animal Health Division. (Location of this office can be ascertained by consulting the publication entitled 'Directory of Consumer Protection Programs, Establishments, Circuits, and Officials'.) Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.4.1.1.2 Establishment sources. Examination shall be made to determine that all processed milk products have been manufactured, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement, or from plants listed with a rating of 90 or more in the document entitled 'Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers', or from plants listed by the U.S. Department of Agriculture in the document titled 'Dairy Plants Surveyed and Approved for USDA Grading Service' and listed for the specific products approved. Nonconformance to the above requirements shall be cause for rejection of the lot or the involved quantity of finished product made therefrom, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.4.1.2 Examination of raw whole milk. Examination and testing of raw milks shall be made to determine compliance with the referenced quality requirements of 3.1.1, and as concerns appearance and odor. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirement(s), reflected by actual examination and tests or by examination of records, shall be cause for rejection of the involved lot of finished product made therefrom.

4.4.1.3 Examination of milk and modified milk product ingredients. Conformance of dairy product ingredients to the requirements of 3.1.2 through 3.1.5 as concerns identity (kind, raw milk supply or applicable grade) shall be determined by examination of pertinent labels, invoices, USDA grade certificate, plant records, or other valid documents reflecting appearance, odor and all other required product quality test data of raw milk used in their manufacture, as specified in 3.1.1. Failure to conform to one or more of the above referenced requirements, reflected by actual examination and testing or by examination of records, shall be cause for rejection of the involved lot finished product.

4.4.1.4 Examination of vegetable fat. Examination of vegetable fat shall be made in accordance with the examination criteria for product characteristics of MIL-F-35081, (note Wiley melting point exception in 3.1.6), except that the sampling plans and acceptable quality levels (AQLs) shall not apply. The sample for examination shall be approximately 1-pound composite derived from 5 primary containers or all containers if less than 5 form a lot. Nonconformance to one or more requirements shall be cause for rejection of the involved lot of finished product.

4.4.1.5 Examination of coconut fat. Conformance of this fat to the requirements of 3.1.7 as concerns identity shall be determined by examination of pertinent labels, invoices or other valid documents. In addition, fat shall be examined organoleptically, as necessary, to determine conformance to the condition requirements as concerns flavor, odor, appearance and presence of extraneous material. The sample for examination shall be approximately 8 ounces taken from each of 5 primary containers or all containers if the lot contain less than 5 containers. Nonconformance to one or more identity or condition requirements shall indicate an unacceptable product and use of such shall be cause for rejection of the involved lot of end item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.4.1.6 Examination of sweeteners, vanilla flavoring, stabilizers, stabilizing salts, emulsifiers, salt and coloring material. Examination of these products shall be made to determine conformance to the requirements of 3.1.8 through 3.1.11, as concerns identity (kind and type). This determination shall be ascertained by examination of labels, invoices or other valid documents. In addition, salt shall be examined for color to determine conformance with the condition requirement in 3.1.11. The sample for examination shall be approximately 8 ounces of ingredient taken from each of 5 primary containers or all containers if the lot contains less than 5 containers. Nonconformance to one or more identity or condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved lot of end item.

4.4.2 Testing of components. Testing of components described in this specification shall be as shown in table III. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE III. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Cans and lids 1/	1 can and 1 lid	Cans	S-1	Tinplate	Nearest 0.01 lb/ base box	4.5.2.3 4.5.2.4 and 5.1.1
Coconut fat	2/	Primary containers	2/	Color	Pass or fail	4.5 and 3.1.7.1
				Moisture	Nearest 0.1 percent	4.5 and 3.1.7.2
				Wiley melting point	Nearest degree Fahrenheit	4.5.2.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE III. Testing of components (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmt. para.
Coconut fat (cont'd)				Peroxide value	Nearest 0.1 meq./kg of fat	4.5.2.1
				Free fatty acids (as oleic acid)	Nearest 0.01 percent	4.5.2.1
				Iodine value	Nearest whole number	4.5.2.1
				Fat stability (A.O.M.)	Pass or fail 3/	4.5.2.1

1/ Exterior can coating, when required, and can label shall be tested in accordance with the applicable subsidiary specifications except that tests shall be performed on the same cans submitted for tinplate testing.

2/ The sample for testing shall be an approximate 1-pound composite derived from 5 primary containers or all containers if less than 5 form a lot.

3/ If failure, report to nearest hour.

4.4.3 Process examination. Unless otherwise specified (see 6.1), examination shall be made during processing to determine compliance with requirements of 3.2 as concerns time, time intervals, temperature, vacuum, when applicable, equipment, procedures and controls. Time and temperature records shall be maintained. Noncompliance with the above requirements, reflected either by examination of records or actual examination, shall be cause for rejection of the involved quantity of finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.4.3.1 Examination of pre-packing storage (when applicable). Examination shall be made to determine compliance with the prepackaging holding requirements of 3.2.4 as concerns storage conditions (use of inert gas or vacuum), storage time before final packaging, vacuum control, dry blending time limitations (when applicable) and type of storage containers. Storage time and vacuum records shall be maintained. Noncompliance with the above referenced requirements as reflected by examination of records or by actual examination shall be cause for rejection of the lot.

4.4.4 Examination of end item. Examination of end item shall be performed in accordance with tables V through VIII and 4.4.4.1 through 4.4.4.5. The examination criteria for the above mentioned tables shall be as shown in table IV and the acceptable quality levels (AQLs) for tables V and VIII shall be expressed as defects per hundred units and for tables VI and VII, as percent defective.

TABLE IV. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs			
				Major A	Major B	Minor	Total
V	S-4	1 filled and sealed can	Cans	0.25	1.5	-	4.0
VI	S-3	1 filled and sealed can	Cans	-	-	2.5	-
VII	S-3	1 filled and sealed can	Cans	-	0.65	-	-
VIII	S-2	Contents of 1 filled can	Cans	1/	2.5	-	-

1/ See footnote 1/ to table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE V. External examination of cans 1/

Category			Defect
Major A	Major B	Minor	
101			Hole or leaker.
102			False seam or improperly closed.
	151		Not type, size or style specified.
	152		Severe dent.
	153		Dent causing sharp ridge.
	154		Cable cut.
	155		Pitted rust. 2/
		201	Moderate dent.
		202	Not clean. 3/

1/ Use USDA - CMS Visual Aids for Inspection of Metal Containers as a guide for classifying can defects.

2/ Rust that can be removed with a soft cloth will not be considered as pitted rust.

3/ Cans showing a very thin film of grease which is discernible to the touch but not readily discernible visually are considered to be clean.

TABLE VI. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight less than 4 lbs., 2.5 ounces 2/

1/ Lot shall be rejected if sample data indicate lot average net weight is less than the specified net weight.

2/ Report results to the nearest 1/4 ounce.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE VII. Examination for can leakage (see 4.5.1.2)

Category	Defect
Major B	
151	Steady progression of bubbles.

TABLE VIII. Examination of product characteristics 1/

Category		Defect
Major A	Major B	Dehydrated mix
101		Presence of foreign flavor or odor, e.g., oxidized, rancid, stale, tallowy, scorched, fishy, etc.
102		Presence of foreign material, e.g., dirt, insect(s), insect part(s), wood, paint, glass, metal particles, etc.
	151	Color not white or light yellow.
	152	Type and style not as specified.
	153	Not uniform in appearance or not well blended.
	154	Presence of lumps which do not fall apart under light pressure.
		Rehydrated mix 2/
	155	Does not reconstitute readily to a typical fluid ice cream mix.
		Freezing performance (see 3.3.2.2) 3/ 4/
	156	Semi-frozen product drawn from freezer lacks a stiff consistency and relatively dry appearance.
	157	Extruded ice cream not characteristically creamy and smooth in texture or lacking in a firm, compact body.
	158	Overrun attainment (batch frozen) not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

TABLE VIII. Examination of product characteristics 1/ (cont'd)

Category		Defect
Major A	Major B	Hardened ice cream 4/
	159	Lacks smooth texture or not free from palpable ice or lactose crystals (sandiness).
	160	Flavor not pleasing or characteristic of vanilla ice cream.

1/ Finding of one or more Major A defects shall be cause for rejection of the lot.

2/ See 3.3.2.1 and 5.4.2.1 for reconstitution instructions.

3/ For purposes of freezing reconstituted powder, select a minimum of 3 primary containers from each lot, disregarding sample size indicated by inspection level S-2 for table VIII cited in table IV.

4/ Sampling and examination for these requirements shall be performed jointly by the Government and the supplier during Government inspection.

4.4.4.1 Examination of exterior can coating (when applicable). Examination of exterior can coating shall be performed in accordance with the applicable examination criteria of TT-C-495.

4.4.4.2 Examination of can labeling. Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.4.4.3 Examination of age of product at time of delivery. Examination shall be made at time of delivery to determine compliance with 3.5. This determination shall be ascertained by examination of records, codes, marked date of pack or other valid means. Noncompliance with the above referenced requirement shall be cause for rejection of the lot.

4.4.4.4 Examination of shipping container. Examination of the filled and closed shipping container shall be made in accordance with the examination criteria contained in the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: Markings missing, incorrect or illegible; Minor: Arrangement not as specified. For level B packs, when applicable, shipping containers reinforced with other than pressure sensitive adhesive filament reinforced tape shall not be acceptable. When packing for level C is required to be in accordance with 5.2.3.1, examination shall be in accordance with the aforementioned specification except that only the defects for marking and arrangement shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

4.4.4.5 Examination of palletized loads. Palletized loads shall be examined in accordance with the criteria of MIL-L-35078.

4.4.5 Sampling procedures and acceptance criteria for testing of finished product. The finished product (dehydrate mix) shall be tested for total fat, protein, moisture, scorched particles, iron, copper (see footnote 1/ to table I), oxygen and microbiological content in accordance with the requirements of 3.3.3 and 3.3.4. Procedures for testing shall be in accordance with 4.5. Test requirements for oxygen and microbiological content shall be applicable to the sample unit. The sample unit for testing shall be one filled and sealed primary container. Lot size shall be expressed in terms of primary containers and the sample size shall be the number of sample units indicated by inspection level S-1. The AQL, expressed as percent defective, and applicable to the oxygen test only shall be 1.5. A defect shall be scored if a test result exceeds 2.0 percent oxygen. The sample unit for testing for fat, protein, moisture, scorched particles, iron and copper, shall be approximately a 1-pound composite derived from the samples used for unit testing. Results shall be reported to the same unit or decimal as specified for the pertinent requirements except that all microbiological counts shall be reported in accordance with Standard Methods for Examination of Dairy Products and the presence of Salmonellae, as positive or negative. One or more test results not conforming to the above referenced requirements, as applicable, shall be cause for rejection of the involved lot of finished product.

4.5 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Examination procedures.

4.5.1.1 Freezing performance examination. The freezing performance of the mix shall be conducted as follows:

(a) Reconstitute the ice cream mix and chill to 36-38° F.

(b) Freeze the chilled mix in an ice cream freezer.

(c) Remove the frozen product into a container and place immediately into a hardening cabinet which has a temperature of less than 0° F. Allow to harden for 24 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

(d) Overrun shall be computed by using the following formula:

$$\frac{A - B}{B} \times 100 = \text{percent overrun}$$

A = weight of unit volume of ice cream mix

B = weight of same volume of frozen ice cream

4.5.1.2 Leakage. The filled and sealed container shall be examined for leakage by submerging in water, contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as leakers.

4.5.2 Test procedures.

4.5.2.1 Chemical analyses. Chemical analyses shall be made in accordance with the methods and publications listed below:

Test	Source	Method
Ingredients:		
Peroxide value	1/	Cd 8-53
Free fatty acids	1/	Ca 5a-40
Moisture	1/	Ca 2e-55
Iodine value	1/	Cd 1-25
Wiley melting point	1/	Cc 2-38
Fat stability (AOM)	1/	Cd 12-57
Color (coconut fat)	1/	Cc 13b-45
Finished product:		
Fat	2/	Fat in dried milk
Protein	2/	Protein
Moisture	3/	Toluene distillation
Iron	4/	Iron
Copper	3/	Copper
Scorched particles	3/	Scorched particles (22g. sample)
Oxygen in headspace gas	3/	Oxygen

1/ Official and Tentative Methods of the American Oil Chemists' Society, Section C, Commercial Fats and Oils.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

2/ Official Methods of Analysis of the Association of Official Analytical Chemists' (A.O.A.C.). Chapter: Dairy Products; Section: Dried Milk, Nonfat Dry Milk, and Malted Milk.

3/ American Dry Milk Institute Bulletin 916, Standards for Grades of Dry Milks, including Methods of Analysis.

4/ A.O.A.C. - Chapter: Cereal Foods; Section: Wheat Flour.

4.5.2.2 Microbiological analyses. Microbiological analyses shall be performed in accordance with the following methods from Standards Methods for the Examination of Dairy Products except as noted:

Test	Chapter	Method
Finished Product:		
Standard plate count	Microbiological Methods for Concentrated Milk and Dry Milk	Agar Plate
Coliform Count	Loc. cit.	Coliform Group (Desoxycholate lactose agar)
Yeast and Mold Count	Loc. cit.	Yeast and Mold count
Salmonella test	1/	

1/ U.S. Department of Health, Education and Welfare, Bacteriological Analytical Manual, Dried Milk Section.

4.5.2.3 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.5.2.4 Test for side seam coating coverage. The side seam area of the can or can body to be tested shall be immersed for 1 to 2 minutes in an electrodeless copper plating solution at room temperature (65 to 90° F.). Area without coating will be indicated by a plating of reddish colored copper. The amount of uncoated area shall be estimated and reported as a percentage of the total side seam area. Prior to use in the test, the plating solution shall be checked for plating adequacy by immersing in it the side seam area of a non-striped can on which it should deposit a readily visible reddish copper plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. Four pounds and 4 ounces of the type and style of ice cream mix specified (see 6.1), shall be filled into a size 603 by 700, open-top style, round, metal can with soldered side seam and compound-lined, double-seamed end. The can shall be made throughout from not less than commercial 0.25-pound electrolytic tin plate per base box (see 4.5.2.3) and shall be coated outside with a coating conforming to type I or, when specified (see 6.1), type III of TT-C-495. Alternatively, the exterior side seam striping specified in TT-C-495 may be clear. When clear striping is used, coverage shall be tested in accordance with 4.5.2.4. The product shall be packaged under an atmosphere of nitrogen.

5.1.2 Level C. The ice cream mix shall be packaged in accordance with 5.1.1, except that cans with or without commercial exterior coating will be acceptable.

5.2 Packing. Six cans of the ice cream mix, arranged 3 in length, 2 in width, and 1 in depth, shall be packed on end in a snug-fitting shipping container in accordance with level A, B, or C, as specified (see 6.1). RSC containers may be top-opening or end-opening.

5.2.1 Level A. The shipping container shall be a fiberboard box, with sleeve, constructed, closed, and reinforced in accordance with style RSC-SL, V2s of PPP-B-636. When unit loading is specified, sleeve and reinforcement is not required.

5.2.2 Level B. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V3c, V3s or V4s, of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with pressure sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

5.2.3.1 When specified (see 6.1 and 6.5), the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

5.3 Unit loads. When specified, (see 6.1) the ice cream mix, packed as specified in 5.2 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1).

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Labeling shall be in accordance with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

5.4.1.2 Shipping containers. Shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military agencies.

5.4.2.1 Unit containers. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use and explanatory statements shall be included:

Directions: Pour contents of can (4-1/4 lb.) on 3-1/2 quarts of water (see caution), mix well, and freeze.

Caution: Water may be warmed to aid in reconstitution. Do not heat powder directly on stove or open flame. Never add warm (over 100° F.) mix to freezer.

NOTE: This product is gas packed.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should exercise the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type and style of ice cream mix required (see 1.2 and 5.1.1).
(c) When examination for prepackaging holding time is required (see 1.2.4).
(d) When in-process examination is not required (see 4.4.3).
(e) When type III exterior can coating is required (see 5.1.1).
(f) Levels of packaging and packing required (see 5.1 and 5.2).
(g) When packing specified in 5.2.2.1 or 5.2.3.1 is required.
(h) Type and class of unit load, when unit loading is specified (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:43Z
Check the source to verify that this is the current version before use.

6.2 Product information. Although refrigeration is not necessary, stability is improved if product is stored in a cool location. Type II product is a dehydrated mix used to make a frozen dessert identical to ice cream except that it contains edible fats of a non-dairy origin in lieu of milk fat. Type II product is intended as a substitute for type I (containing milk fat), when authorized, in areas where its manufacture and sale is not prohibited by law.

6.3 (Military agencies). Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15.

6.4 (Military agencies). Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific overseas requirements.

6.5 (Military agencies). Packing specified in 5.2.3.1 is intended for direct shipment from the supplier to the first receiving activity for immediate use within CONUS.

6.6 This specification includes the requirements of Military specifications MIL-I-00705D(GL) dated 18 December 1969 and MIL-I-705C dated 17 August 1965.

6.7 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only. Unless otherwise specified, such inspection should be made by the receiving facility.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

VA DMS
HEW - FDA, NIH
USDA - CMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 20 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:43Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Ice Cream Mixes, Regular And Imitation, Dehydrated EE-I-1699

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:43Z
Check the source to verify that this is the currentversion before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:43Z
Check the source to verify that this is the current version before use.